

Cassandra PPT.pptx

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Summary

Apache Cassandra is a **highly scalable, column-oriented database** that originated at Facebook and evolved into an Apache project, designed to handle massive data loads across a **distributed peer-to-peer network**. By utilizing a **masterless architecture** and robust **replication strategies**, the system avoids a single point of failure and ensures data remains accessible even during node outages. The text highlights its unique data model—composed of **clusters, keyspaces, and column families**—and emphasizes its **tunable consistency**, which allows users to balance speed and accuracy during read and write operations. Finally, the source provides a practical guide to **Cassandra Query Language (CQL)**, illustrating how to manage data through commands for creation, indexing, and the use of **specialized collection types** like sets, lists, and maps.

Keywords: Apache Cassandra Overview, Replication Strategy, Data Consistency Levels, Cassandra Data Model, CQL Query Operations

Content

Apache Cassandra was born at Facebook. After Facebook open sourced the code in 2008, Cassandra became an Apache Incubator project in 2009 and subsequently became a top-level Apache project in 2010. It is a column-oriented database designed to support peer-to-peer symmetric nodes instead of the master-slave architecture. It is built on Amazon's dynamo and Google's BigTable. It is highly scalable, high performance distributed database. It is column oriented database designed to support peer to peer network. Adherence to availability and partition tolerance of CAP. Apache Cassandra - An Introduction. Big Data and Analytics by Seema Acharya and Subhashini Chellappan. Copyright 2015, WILEY INDIA PVT. LTD. Features: Peer to Peer Network-Designed to distribute and manage large data loads across multiple nodes in a cluster constituted of commodity hardware-Does NOT have a master slave architecture-NOT have a single point of failure-Graceful degradation where everything does not come crashing at any instant owing a node

failureEnsures data is distributed across all nodes in the cluster Each node exchange information across the cluster every second2. Gossip and Failure Detection Gossip protocol is used for intra ring communicationPeer to peer communication protocol which eases the discovery and sharing of location and state information with other nodes in the clusterA node only has to send out the communication to a subset of other nodes 3. Partitioner A partioner takes a call on how to distribute data on the various nodes in the clusterIt also determines the node on which to place the very first copy of the dataA partioner is a hash function is used to compute the token of the partion key4. Replication factor rtiNode Node is the place where data is stored. It is the basic component of Cassandra.Data Center A collection of nodes are called data center. Many nodes are categorized as a data center.Cluster The cluster is the collection of many data centers.As hardware problem can occur or link can be down at any time during data process, a solution is required to provide a backup when the problem has occurred. So data is replicated for assuring no single point of failure.Cassandra places replicas of data on different nodes based on these two factors.1. Where to place next replica is determined by the Replication Strategy. 2.While the total number of replicas placed on different nodes is determined by the Replication Factor.One Replication factor means that there is only a single copy of data while three replication factor means that there are three copies of the data on three different nodes.For ensuring there is no single point of failure, replication factor must be three.There are two kinds of replication strategies in Cassandra.1. SimpleStrategySimpleStrategy is used when you have just one data center. SimpleStrategy places the first replica on the node selected by the partitioner. After that, remaining replicas are placed in clockwise direction in the Node ring.Here is the pictorial representation of the SimpleStrategy.There are two kinds of replication strategies in Cassandra.2. NetworkTopologyStrategyNetworkTopologyStrategy is used when you have more than two data centers.In NetworkTopologyStrategy, replicas are set for each data center separately. NetworkTopologyStrategy places replicas in the clockwise direction in the ring until reaches the first node in another rack.This strategy tries to place replicas on different racks in the same data center. This is due to the reason that sometimes failure or problem can occur in the rack. Then replicas on other nodes can provide data.Here is the pictorial representation of the Network topology strategyRead consistency means how many replicas must respond before sending out the result to client applicationWrite consistency is how many replicas write must succeed before sending out an acknowledgement to the client application5. Anti Entropy and Read RepairA client can connect to any node in the cluster to read dataHow many nodes will be read before responding to the client is based on the Consistency level specified by the client If few of the nodes respond with an out of date value Cassandra will initiate a read repair to bring the replicas with stale values up to dateThis is done using Anti Entropy gossip protocol.Anti entropy implies comparing all the replicas of each piece of data and updating each

replica to the newest version. 6. Write operation in Cassandra When write request comes to the node, first of all, it logs in the commit log. Commit log is used for crash recovery. After data written in Commit log, data is written in Mem-table. Data written in the mem-table on each write request also writes in commit log separately. Mem-table is a temporarily stored data in the memory while Commit log logs the transaction records for back up purposes or crash recovery. When mem-table is full, data is flushed to the SSTable data file. 7. Hinted Handoffs Client BC Replicates Row K Coordinator Node C is down. Write a hint in your table A. Writes Row K. System hints table. Three nodes A, B, C. C is down. Replication factor is 2. Write operation on node A which is the coordinator and serves as a proxy. When row k is written by the client to Node A, it will write row K to node B and stores a hint for Node C. Hint has the following information: Location of the node on which replica is to be placed, Version, Mtada, Actual data. 8. Tunable Consistency Strong Consistency- Each update propagates to all location where that piece of data resides. Eventual Consistency- Client is acknowledged with success as soon as part of the cluster acknowledges the write. Big Data and Analytics by Seema Acharya and Subhashini Chellappan Copyright 2015, WILEY INDIA PVT. LTD. Read Consistency Big Data and Analytics by Seema Acharya and Subhashini Chellappan Copyright 2015, WILEY INDIA PVT. LTD. ONE Returns a response from the closest node (replica) holding the data. QUORUM Returns a result from a quorum of servers with the most recent timestamp for the data. LOCAL_QUORUM Returns a result from a quorum of servers with the most recent timestamp for the data in the same data center as the coordinator node. EACH_QUORUM Returns a result from a quorum of servers with the most recent timestamp in all data centers. ALL This provides the highest level of consistency of all levels and the lowest level of availability of all levels. It responds to a read request from a client after all the replica nodes have responded. Big Data and Analytics by Seema Acharya and Subhashini Chellappan Copyright 2015, WILEY INDIA PVT. LTD. Write Consistency ALL This is the highest level of consistency of all levels as it necessitates that a write must be written to the commit log and Memtable on all replica nodes in the cluster. EACH_QUORUM A write must be written to the commit log and Memtable on a quorum of replica nodes in all data centers. QUORUM A write must be written to the commit log and Memtable on a quorum of replica nodes. LOCAL_QUORUM A write must be written to the commit log and Memtable on a quorum of replica nodes in the same data center as the coordinator node. This is to avoid latency of inter-data center communication. ONE A write must be written to the commit log and Memtable of at least one replica node. TWO A write must be written to the commit log and Memtable of at least two replica nodes. THREE A write must be written to the commit log and Memtable of at least three replica nodes. LOCAL_ONE A write must be sent to, and successfully acknowledged by, at least one replica node in the local data center. Cassandra - Data Model The data model of Cassandra is significantly different from what we normally

see in an RDBMS. ClusterCassandra database is distributed over several machines that operate together. The outermost container is known as the Cluster. For failure handling, every node contains a replica, and in case of a failure, the replica takes charge. Cassandra arranges the nodes in a cluster, in a ring format, and assigns data to them. KeyspaceKeyspace is the outermost container for data in Cassandra. The basic attributes of a Keyspace in Cassandra are –Replication factor – It is the number of machines in the cluster that will receive copies of the same data.Replica placement strategy – It is nothing but the strategy to place replicas in the ring. We have strategies such as simple strategy (rack-aware strategy), old network topology strategy (rack-aware strategy), and network topology strategy (datacenter-shared strategy).Column families – Keyspace is a container for a list of one or more column families. A column family, in turn, is a container of a collection of rows. Each row contains ordered columns. Column families represent the structure of your data. Each keyspace has at least one and often many column families.The syntax of creating a Keyspace is as follows –CREATE KEYSPACE Keyspace name WITH replication = {'class': 'SimpleStrategy', 'replication_factor' : 3}; The following illustration shows a schematic view of a Keyspace. Column FamilyA column family is a container for an ordered collection of rows. Each row, in turn, is an ordered collection of columnsThe following figure shows an example of a Cassandra column family. RDBMSCassandraRDBMS deals with structured data.Cassandra deals with unstructured data.It has a fixed schema.Cassandra has a flexible schema.In RDBMS, a table is an array of arrays. (ROW x COLUMN)In Cassandra, a table is a list of “nested key-value pairs”. (ROW x COLUMN key x COLUMN value)Database is the outermost container that contains data corresponding to an application.Keyspace is the outermost container that contains data corresponding to an application.Tables are the entities of a database.Tables or column families are the entity of a keyspace.Row is an individual record in RDBMS.Row is a unit of replication in Cassandra.Column represents the attributes of a relation.Column is a unit of storage in Cassandra.RDBMS supports the concepts of foreign keys, joins.Relationships are represented using collections.The following table lists down the points that differentiate the data model of Cassandra from that of an RDBMS.

RDBMS	Cassandra
10:001,12:002,11:003,22:004;Smith:001,Jones:002,Johnson:003,Jones:004;Joe:001,Mary:002,Cathy:003,Bob:004;60000:001,80000:002,94000:003,55000:004;Set	What Cassandra does not support
There are following limitations in Cassandra query language (CQL).CQL does not support aggregation queries like max, min, avgCQL does not support group by, having queries.CQL does not support joins.CQL does not support OR queries.CQL does not support wildcard queries.CQL does not support Union, Intersection queries.Table columns cannot be filtered without creating the index.Greater than (>) and less than (<) query is only supported on clustering column.Cassandra query language is not suitable for analytics purposes because it has so many	

limitations. Create keyspace University Create keyspace University with replication={ 'class':SimpleStrategy, 'replication_factor': 3 }; Create Student Table Student Rollno, Name, Dept Alter table to add semester 'Alter Table' that will add new column in the table Student. Describe the table contents Insert the values into table and update student name of rollno 18 Delete the contents of rollno 1 Select all CS dept students Remove index Create Teacher Table- Id, Name, Email Set collection that store multiple email addresses for the teacher. Add Courses to Teacher table insert in column "coursenames". shows the current database state after insertion. Write query when you want to save course name with its prerequisite course name data is being inserted in map collection type Remove the contents of table Drop table and Keyspace Drop keyspace University;